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TECHNICAL ADVANCE

Primed *in situ* labelling facilitates flow sorting of similar sized chromosomes

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Summary

Flow karyotyping and sorting of individual chromosome types is difficult when chromosomes of a complement do not differ sufficiently in DNA content. A strategy for sorting chromosomes of similar size has been developed. For this purpose oligonucleotide primed *in situ* (PRINS)-labelling was adapted to field bean chromosomes in suspension. With a primer designed according to a tandemly repetitive sequence (*FokI* element) PRINS-labelling resulted in fluorescence signals specific in position and intensity for each chromosome. A bivariate sorting mode combining fluorescence pulse areas obtained from propidium iodide staining (representing DNA content) and fluorescein isothiocyanate signals (representing chromosome-specific label) allowed chromosomes deviating in length by less than 1% of the haploid metaphase complement to be sorted. The average purity of sorted fractions was 95%. This technique should be applicable also to chromosomes of other species for obtaining chromosome-specific painting probes, for construction of chromosome-specific libraries (both without additional DNA amplification), and for gene mapping.

Introduction

Flow cytometry is a powerful tool for rapid and precise karyotyping, for detection of numerical and structural chromosome aberrations, and for sorting of chromosomes (for review see Doležel *et al.*, 1994; Gray and Cram, 1990). Large-scale sorting of individual chromosome types may be used to establish DNA probes for chromosome painting, to construct chromosome-specific DNA libraries (both without additional amplification steps which are necessary when microdissected chromosomes are used (Pich *et al.*, 1994) and may cause a bias between sequence classes), and to map genes (Collard *et al.*, 1985; Cotter *et al.* 1989; Macas *et al.*, 1993). It requires pure chromosome suspensions prepared from synchronously dividing cells and sufficient differences in size (DNA content) of the chromosomes to be sorted.

Flow sorting of chromosomes is reported for a few plant species: tomato (Arumuganathan *et al.*,1991, 1994), wheat (Wang *et al.*, 1992), *Haplopappus gracilis* (De Laat and Blaas, 1984; De Laat and Schel, 1986), *Petunia hybrida* (Conia *et al.*, 1988), *Melandrium album* (Veuskens *et al.*,1992), and *Vicia faba* (Lucretti *et al.*, 1993; Macas *et al.*,1993). Except for *H. gracilis* ($n=2$) and reconstructed karyotypes of *V. faba*, it was not possible to sort each chromosome type of the complement. This was due to insufficient differences in chromosome size. Even the application of bivariate flow karyotyping, using a second fluorochrome with preferential affinity to AT or GC base pairs (Arumuganathan *et al.*, 1991; Conia *et al.* 1988; Wang *et al.*, 1992) did not significantly improve the discrimination of similar chromosome types. Therefore, a second fluorescent marker, not correlated with the total chromosomal DNA content is desirable.

Here we describe oligonucleotide-primed *in situ* (PRINS)-labelling (Koch *et al.*, 1989) on plant chromosomes in suspension. PRINS-labelling of a tandemly repeated sequence (*FokI* element, Kato *et al.*, 1984) allowed the sorting of all six chromosomes of a reconstructed karyotype (ACB) of the field bean of which only two chromosomes could be sorted on the basis of their DNA content.

Results and Discussion

Chromosome-specific PRINS-labelling of field bean chromosomes

At first the specificity of the PRINS-technique was tested on field bean chromosomes of squash preparations using a primer designed according to the *FokI* element sequence. The *in situ* extension of this primer by Taq polymerase and detection of incorporated DIG-dUTP by anti-DIG-FITC yielded strong signals at all chromosome loci which

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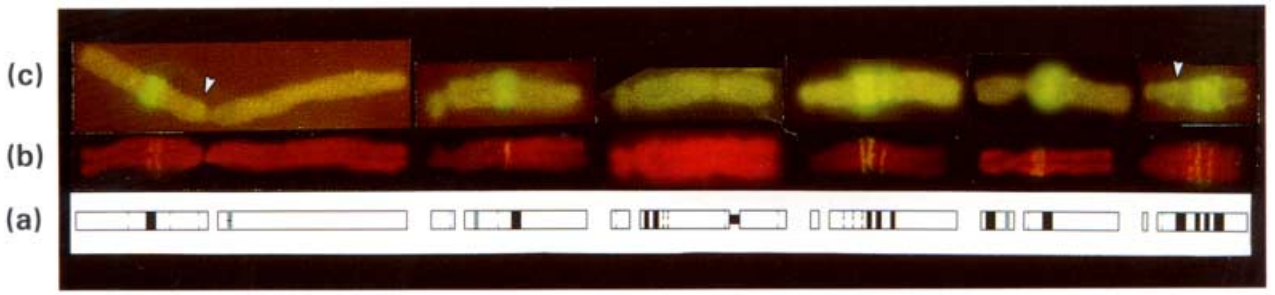


Figure 1. Labelling of heterochromatic *FokI* element loci on chromosomes of the field bean karyotype ACB.
(a) Scheme of Giemsa-banded chromosomes of the reconstructed karyotype ACB. Karyotype ACB is characterized by two reciprocal translocations between chromosomes I and III, and I and VI, respectively. In addition, chromosome V of this karyotype is involved in a pericentric inversion (see Döbel *et al.* 1978).
(b) Signals after FISH with biotinylated *FokI* sequence repeats.
(c) Signals obtained after PRINS-labelling with one *FokI* element-specific primer and detection of incorporated DIG-dUTP by anti-DIG-FITC. In addition to the signals obtained by FISH, two minor signals on chromosome I and VI (arrows) were found.

revealed signals after fluorescent *in situ* hybridization (FISH) with biotinylated strands of the *FokI* sequence element as a probe (see Figure 1 and Fuchs *et al.*, 1994). Even two minor signals, not regularly detectable after FISH, were additionally found at positions of low-frequency Giemsa bands in the short arm of chromosome I and the long arm of chromosome VI. The same result was obtained by PRINS-labelling of chromosomes in suspension. The position of the signals allowed us to distinguish each chromosome microscopically, independent of other morphological criteria. The total intensity of signals obtained for each chromosome was mostly uncorrelated with the total DNA content of the corresponding chromosomes (Figure 2). The more or less uniform fluorescence labelling outside the *FokI* element loci observed after PRINS-labelling probably is due to the background activity of Taq polymerase at 3'-ends of abundant nicks and gaps along the chromosomes. This conclusion is supported by the observation that DNA of micro-isolated reconstructed chromosomes, which showed no signals after FISH with *FokI* sequence probes, yielded no amplification products after PCR with *FokI* element-specific primers (Fuchs *et al.*, 1994; Pich *et al.*, 1994,). While such background fluorescence would prevent the detection of single-copy sequences it does not disturb localization of highly repetitive tandem repeats.

Flow karyotyping and sorting of chromosomes

Univariate analyses of the ACB karyotype were based on propidium iodide (PI) and fluorescein isothiocyanate (FITC) fluorescence signals, respectively. Signals obtained by PI, an intercalating dye, are strongly correlated with the DNA content of the chromosomes. The typical histogram resulted in two major peaks and some smaller peaks. As evident from chromosome sorting, the first high peak with the smallest chromosome-specific PI fluorescence represents chromosome VI, which is the smallest chromosome of karyotype ACB. Chromosomes II, III, IV, V and the short arm of chromosome I (resulting from mechanical disruption at the centromere of the metacentric chromosome I during the isolation procedure) are combined within the second high peak. The following relatively undifferentiated flat peaks correspond to the long arms of chromosome I and the intact chromosomes I, the largest components of karyotype ACB (Figure 2a).

The FITC histogram (Figure 2b) deviates from that obtained by PI. Chromosomes III, without *FokI* element-specific FITC signals, form the most narrow peak (after those formed by chromosome debris) with the smallest chromosome-specific FITC fluorescence. This peak is overlapped by the major peak representing chromosomes II, VI and both arms of chromosome I. The following broad peak corresponds to chromosomes IV and V with similar FITC signals. Chromosomes I form the next flat peak.

Only bivariate analysis using PI versus FITC fluorescence resolved distinct areas for all six chromosomes of karyotype ACB and the broken arms of chromosome I in a two-dimensional diagram (Figure 2c). The peaks corresponding to single chromosome types or groups of chromosomes in the PI histogram were focused by means of the FITC value in the dot plot. The assignment of sorted chromosomes to distinct regions of the dot plot (Figure 2c) was confirmed by microscopic evaluation of sorted fractions (Figure 3).

The purity of sorted fractions was on average 95% (± 2.9%). Chromosomes II were obtained with 96.5%, chromosomes III with 97%, and chromosomes VI with 96.4% purity. The highest purity (98.6%) was obtained for chromosome V, the lowest for chromosomes I (91%) and IV (92.3%). The fraction of chromosomes I was occasionally contaminated by two acrocentric chromosomes sticking together. The few contaminations within the fraction of chromosomes IV were chromosomes V.

Previously, a bivariate sorting mode considering fluorescence pulse area and fluorescence pulse width allowed us to sort the metacentric chromosome I of the standard karyotype of the field bean from the five acrocentrics which

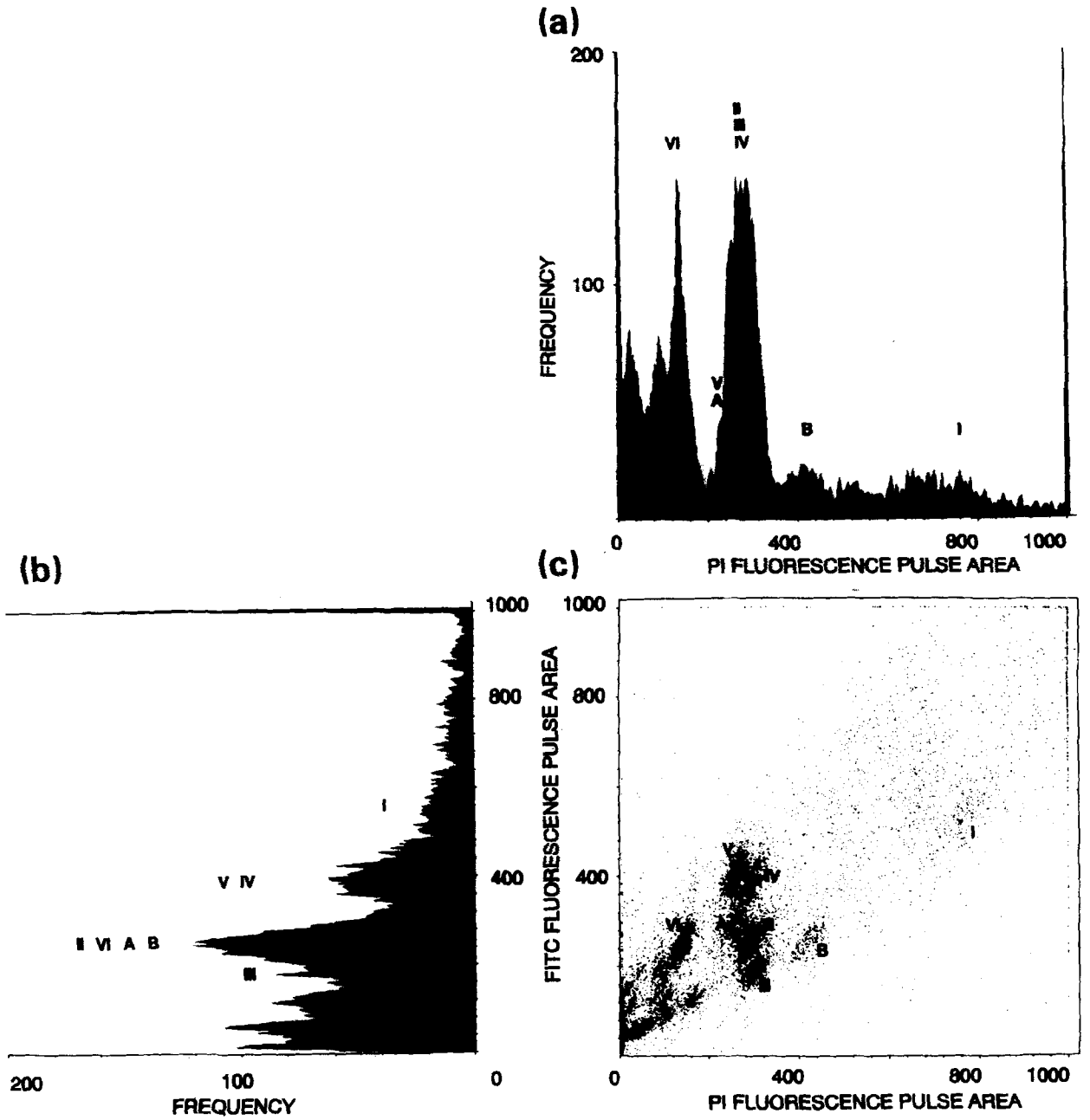


Figure 2 Uni- and bivariate karyotyping of chromosomes of field bean karyotype ACB.
Karyotyping was done on the basis of fluorescence pulse areas. For univariate karyotyping (a,b), the frequency of counted particles is given on the ordinate. Intensities on the abscissae are as arbitrary units expressed on a 1024 channel scale. The peaks corresponding to chromosomes I to VI and separate short (A) and long (B) arms of chromosome I are indicated in (a) and (b).
(a) Univariate flow karyotype obtained after the analysis of PI-stained chromosomes.
(b) Univariate flow karyotype obtained after the analysis of *FokI* sequence-specific FITC signals.
(c) Bivariate flow karyotype (PI versus FITC fluorescence). All chromosomes of the complement (denoted as in (a) and (b)) are clearly resolvable in separate areas.

could not be differentiated from each other (Lucretti *et al.*, 1993). The same sorting mode allowed us to sort five of the six chromosomes of the reconstructed field bean karyotype EFK, with chromosomes largely differing in size, at comparably high purity (Lucretti *et al.*, 1993). However, even a bivariate sorting mode using two fluorochromes with preferential affinity to AT or GC base pairs (DAPI, mithramycin) did not allow us to sort chromosomes II, III, IV and V of karyotype ACB efficiently (data not shown). Thus, only a bivariate sorting mode including fluorescence

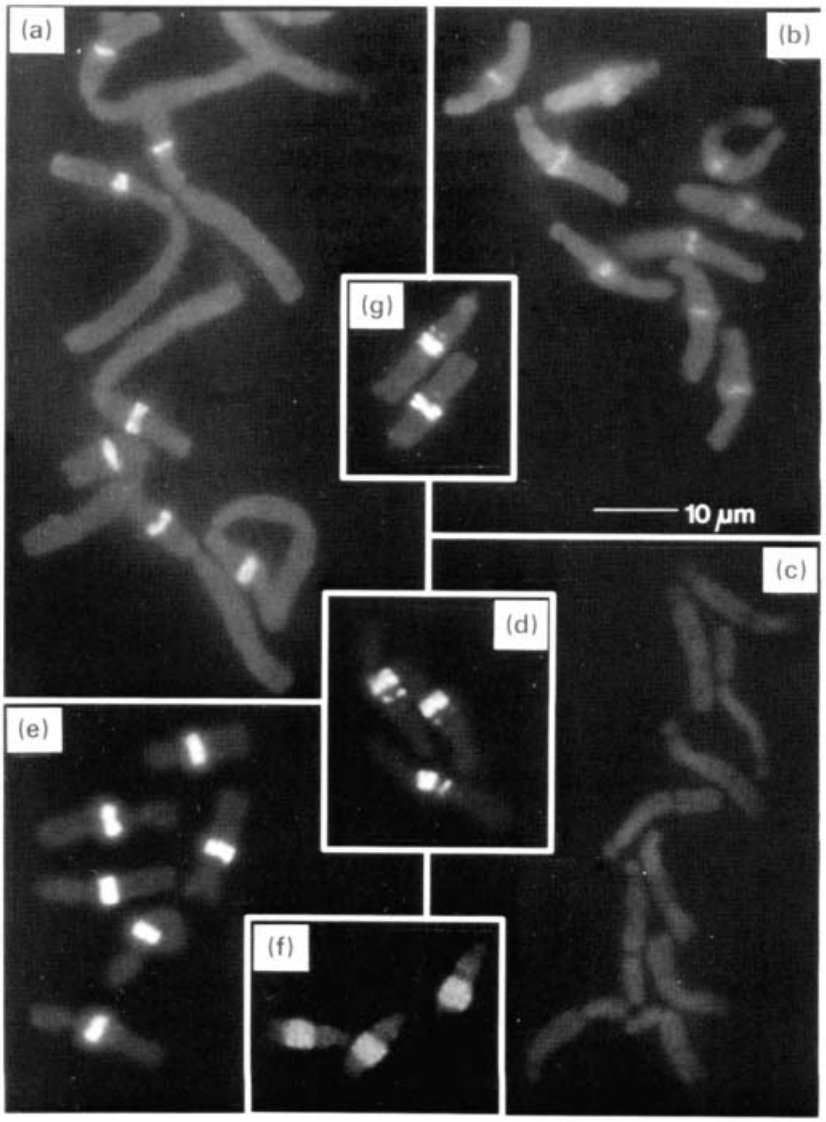


Figure 3 Flow-sorted chromosomes of karyotype ACB after PRINS-labelling of *FokI* element loci on chromosomes in suspension. (a) Chromosomes I, (b): II, (c): III, (d): IV, (e): V, (f): VI, and (g): short arms of chromosome I. Photographs were taken directly on Kodak film (ISO 400).

of the fluorochrome used for detection of PRINS-labelled tandemly repeated sequences allows distinctive sorting of chromosomes as similar in size as chromosomes II, III, IV, V and the broken arms of chromosome I of field bean karyotype ACB which represent proportions of 15.3%, 15.7%, 14%, 12.4%, 13%, and 16.6% of the metaphase chromosome complement, respectively (Michaelis and Rieger, 1971). PRINS-labelling seems to be easier than multistep *in situ* hybridization with chromosomes in suspension (Dudin *et al.*, 1987) and offers more targets than chromosome-specific antibodies that could be considered as alternatives for chromosome-specific labelling. Therefore, PRINS-labelling using oligonucleotide primers designed according to satellite-, rDNA- or 5S RNA- specific sequences should be suitable to sort specific chromosomes out of homogeneous complements of various plant species. In addition to the procedure described here, fluorescence pulse width as a third parameter (Lucretti *et al.*, 1993) and compensation for emissions of PI into the FITC photomultiplier and vice versa could be tested for improvement of chromosome sorting if necessary.

Experimental procedures

Preparation of field bean chromosomes

Main roots (~2 cm) of seedlings of a field bean line with the reconstructed karyotype ACB (for description see Döbel *et al.*, 1978) were treated for 2–3 h in 0.05% colchicine, fixed in 3:1 ethanol:acetic acid for 1 h, washed in sodium-phosphate buffer, pH 4.5, for 1 h, and digested in 1% pectinase (Sigma) and 1% cellulase (Onuzuka R10) for 1 h at 37°C. Meristems were squashed in 45% acetic acid. Slides were used immediately for PRINS or stored in 96% ethanol at −20°C. Synchronization of root tip meristems and preparation of metaphase chromosome suspensions was done as described by Schubert et al. (1993).

PRINS-labelling of chromosomes on slides and in suspension

Squash preparations on slides were denatured in 70% formamide, 2xSSC at 70°C for 10 min, and washed in ice-cold 70% and 96% ethanol, twice for 5 min each. PRINS mix contained 1xPCR buffer (Biomaster, 16 mM $(NH_4)_2SO_4$, 67 mM Tris–HCl (pH 8.8 at 25°C), 0.01% Tween 20), 2.5 mM $MgCl_2$, 420 nM *FokI* sequence-specific primer (5' GCC TTG TCA TTA TGG AAG GTA GTC TG 3', Kato *et al.*, 1984), 1x DIG DNA labelling mix (Boehringer, 0.1 mmol l^{-1} dATP, dCTP, dGTP, each and 0.065 mmol l^{-1} dTTP, 0.035 mmol DIG-dUTP, pH 7.5 at 20°C), and 5U Taq polymerase per 100 µl. A mix of 20 µl was dropped on each prewarmed (37°C) air-dried slide and the cover slip was sealed with fixogum rubber cement. Slides were incubated in a wet chamber (Zytotherm, Schutron) for 5 min at 55°C and for 30 min at 70°C. Reaction was stopped by washing slides for 2 min at 70°C in 50 mM NaCl, 50 mM EDTA (pH 8.0) and twice for 5 min in 4xSSC, 0.1% Tween 20 at room temperature. Thereafter slides were incubated in blocking solution (3% BSA, 4xSSC, 0.1% Tween 20, Lichter and Cremer, 1992) at 37°C for 30 min. Detection was done with anti-DIG-FITC (Boehringer, 20 ng μl^{-1} in 1% BSA, 4xSSC, 0.1% Tween 20, Lichter and Cremer, 1992) at 37°C for 30 min.

For flow karyotyping about 50 000 chromosomes in ~100 µl of isolation buffer were centrifuged at 1200 ***g*** for 15 min. The chromosomes were resuspended in PRINS mix (for concentrations see above) and overlaid with mineral oil. Samples were kept at 94°C for 10 min in a thermal cycler (Perkin Elmer) and then cooled to 85°C. Five units of Taq polymerase per 100 µl mix were added, followed by primer annealing at 50°C for 5 min and primer extension at 70°C for 30 min. The reaction was stopped by adding 1/20 vol of 0.25 M EDTA. Chromosomes were centrifuged again (1200 g,15 min), and resuspended in 50 µl blocking solution at 37°C for 30 min.

Thereafter 10 µl of 200 ng μl^{-1} anti-DIG-FITC (Boehringer, Fab fragments from sheep) were added and incubated again at 37°C for 30 min, or alternatively, at 4°C overnight. After incubation samples were centrifuged (1200 ***g***, 15 min), chromosomes were resuspended in approximately 1 ml of 1xTE and counterstained with PI (50 µg ml^{-1}, final concentration) before flow sorting.

Flow sorting of FITC-labelled chromosomes

Chromosome suspensions were analysed with a FACStar^{Plus} flow cytometer and sorter (Becton & Dickinson) using the macrosort accessory with a 100 µm nozzle. The PI as well as the FITC fluorescence were excited with a 5 W argon-ion laser (Coherent INNOVA 90–5) at 488 nm with 500 mW. Both signals were separated by a dichroic mirror DM 610, transmitting light below and reflecting above 610 nm. The spectral purity of both types of fluorescence was additionally increased by a band-pass filter DF 530 for FITC and a long-pass filter RG 610 for PI fluorescence. High-sensitive photomultipliers R1477 (Hamamatsu) were used for fluorescence detection in both channels. The data acquisition was triggered by the PI fluorescence pulse height signal. For actual flow sorting the signals were taken as fluorescence pulse area which is less sensitive to variations in the degree of chromosome contraction than pulse height. For sorting decisions the linear dot plot of PI versus FITC fluorescence was used.

The instrument was aligned using 2 µm Fluoresbrite Calibration Grade Microspheres (Polyscience), which gave a coefficient of variation (CV) below 2% for the FITC channel. Samples were analysed with a rate of 30–40 particles per second.

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